Finding the Right Law, in Khmer

Can a model match everyday Khmer questions with the correct legal articles?

EXAMPLE USER QUESTION

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

1,976	4	200
LEGAL ARTICLES	NEURAL APPROACHES	NATURAL TEST QUESTIONS

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Why Normal Search Struggles

"A keyword match is not always a meaning match"

តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?

BM25 / KEYWORD SEARCH

Matches words, not meaning

Ranks articles by keyword overlap. It may miss the correct law when the question uses different wording.

NEURAL SEMANTIC RETRIEVAL

Matches meaning, not just words

Converts questions and articles into embeddings, then ranks them by semantic similarity.

Goal: retrieve the relevant article, not generate a new law.

Preparing the Dataset

First, we had to make the Khmer PDFs readable

RAW EXTRACTED TEXT

maRta 336>- karbegáltkic©snña...

AFTER LIMON → UNICODE CONVERSION

មាត្រា ៣៣៦.- ការបង្កើតកិច្ចសន្យា...

Official PDFs
Source documents

→

Convert Limon
Legacy font mapping

→

Normalize Khmer
Character ordering

→

Split into articles
Preserve metadata

1,304 Civil Code articles

672 Criminal Code articles

Creating Training Data

We had legal articles, but not enough question–answer pairs

WEAK SUPERVISION: A TRAINING PAIR

Article title → Article body

We removed titles from article bodies to avoid trivial text matching.

1,176	147	148
training pairs	validation pairs	held-out title queries

Separate main benchmark: 200 natural Khmer questions · 261 reference articles excluded from training.

The Four Models

Four ways to teach a model to retrieve legal articles

A1 · START FROM SCRATCH

BiLSTM

Khmer word tokens

↓

1-layer BiLSTM

↓

512-d vector

1.97M trainable

A2 · FREEZE THE BACKBONE

XLM-R probe

Pretrained XLM-R

↓

Frozen backbone

↓

768-d vector

0.59M trainable

A3 · ADAPT ALL LAYERS

XLM-R fine-tune

Pretrained XLM-R

↓

12 transformer layers

↓

768-d vector

278.04M trainable

A4 · KHMER PRETRAINING

PrahokBART

Khmer pretrained BART

↓

Encoder + pooling

↓

512-d vector

35.83M trainable

How the Models Learn

Contrastive learning: bring matching pairs closer



One correct pair per row

Other article sin the batch act as negative examples.

A3 batch size: 16 · 15 in-batch negatives

Similarity → InfoNCE loss (τ = 0.05) → Backpropagation → Updated model weights

Experimental Setup

Experimental setup & compute

FOUR TRAINED APPROACHES

APPROACH	TRAINABLE	HARDWARE
A1 · BiLSTM	1.97M	CPU
A2 · Linear probe	0.59M	CPU
A3 · Full fine-tune	**278.04M**	**T4 GPU**
A4 · PrahokBART	35.83M	CPU (14 threads)

A3 HYPERPARAMETER SEARCH

6 training runs

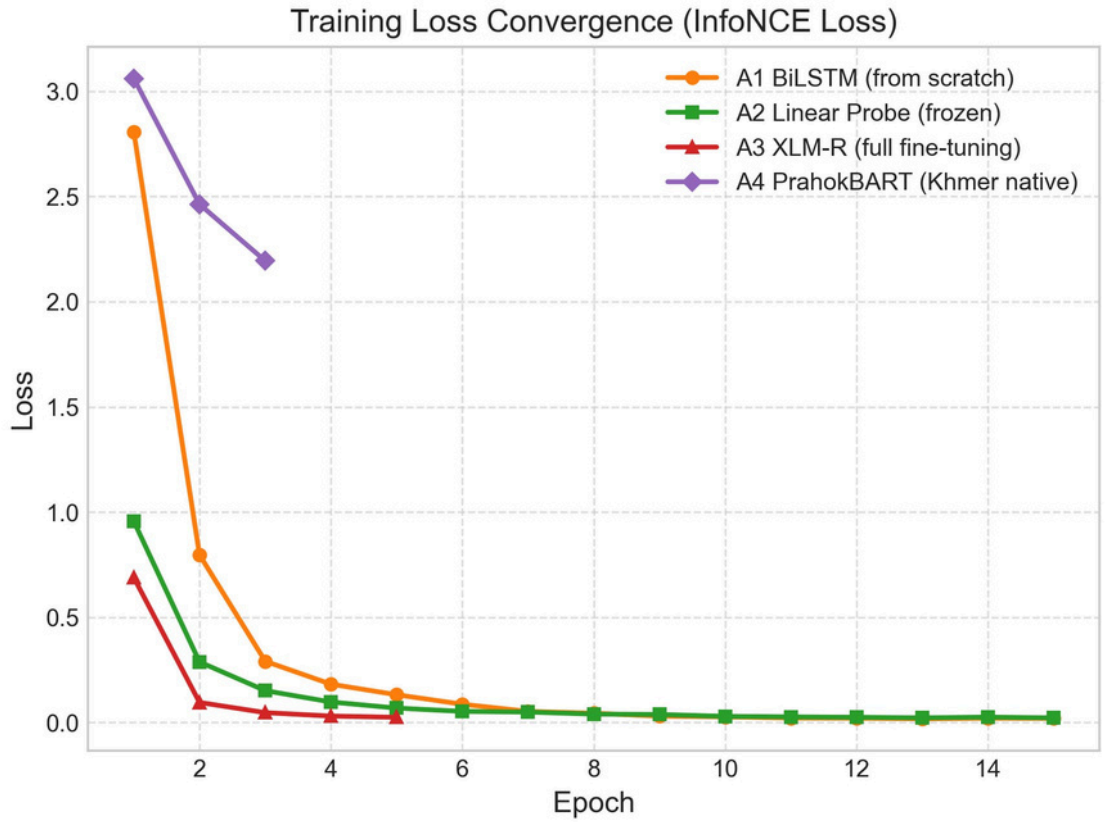
3 learning rates × 2 weight decays

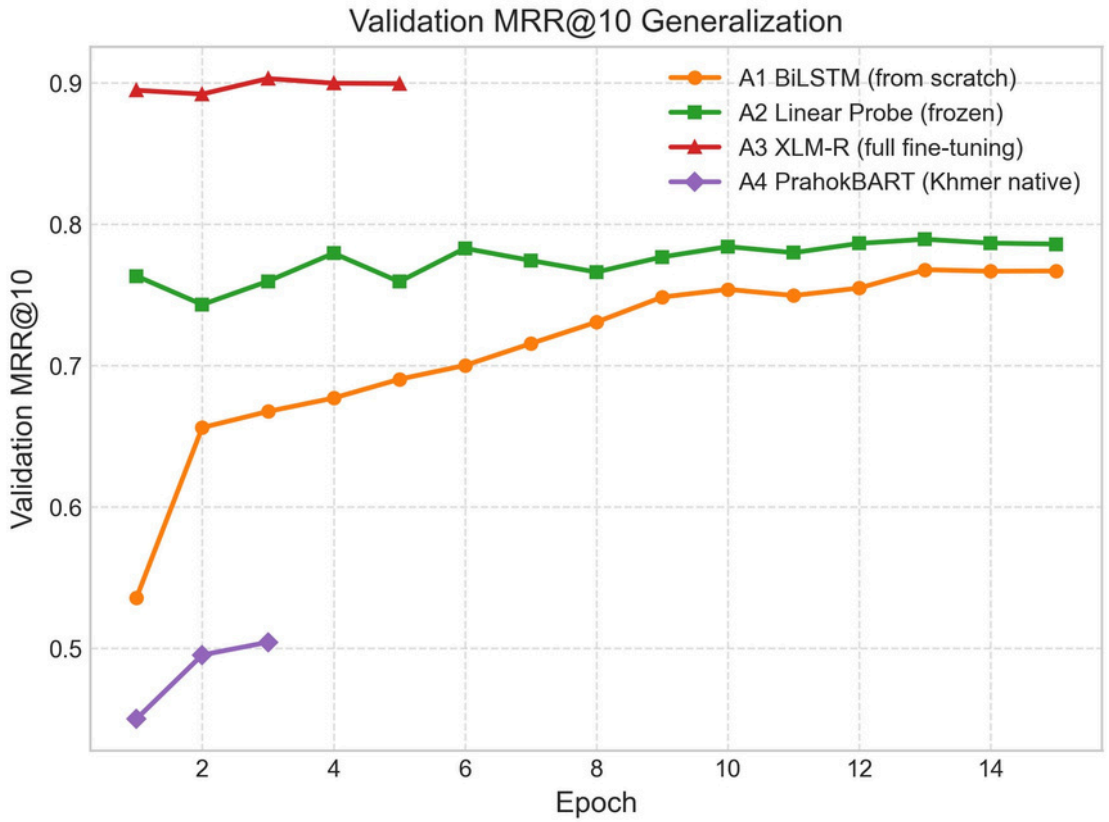
SELECTED CONFIGURATION

2e-5	0.01
learning rate	Weight decay

VAL MRR@10 **0.9095**

What Happened During Training

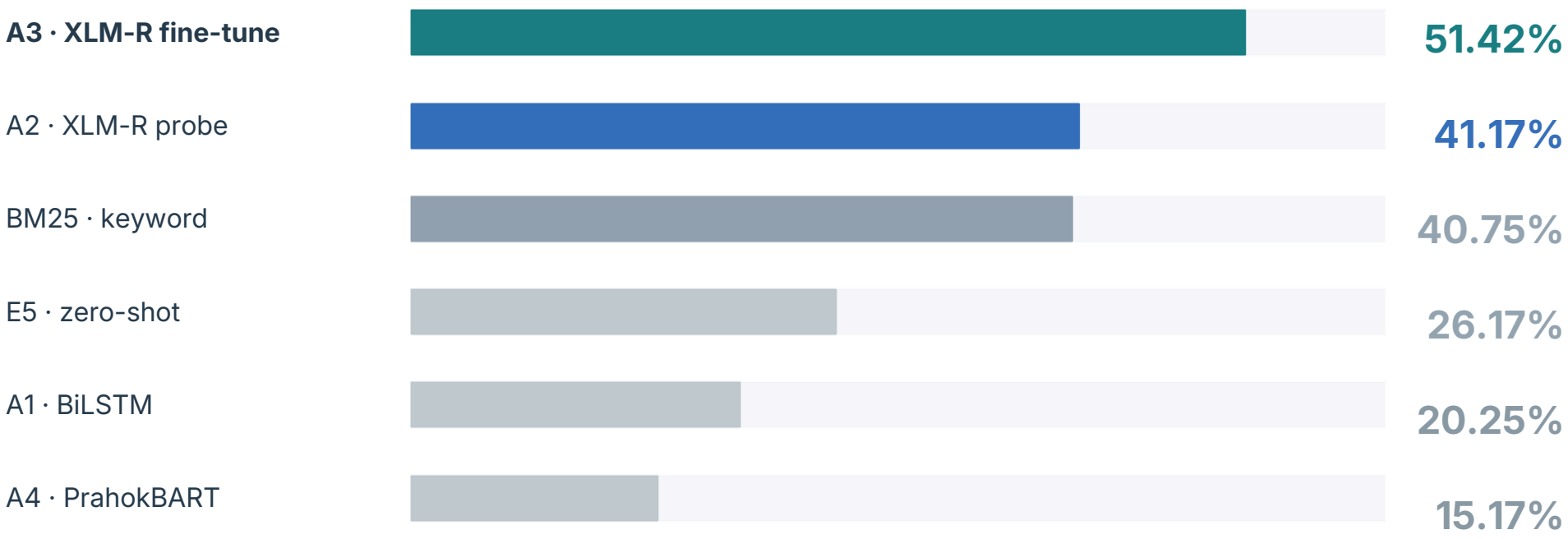




A3: 6 LR/weight-decay runs · selected LR 2×10^{-5}, WD 0.01 · validation MRR@10 0.9095 · Tesla T4

Consolidated Results

RECALL@5 — SHARE OF RELEVANT ARTICLES IN TOP FIVE



A3 VS BM25

+10.67

pp Recall@5

+44.1%

relative MRR

MRR@10: A3 0.4827 · BM25 0.3350 | Recall@5 difference: 10.67 percentage points

Results on the same 200 Khmer questions

All 1,976 articles are candidates. No statute filter is applied in the benchmark.

System	Recall@1	Recall@5	MRR@10	Hit@5
BM25	0.2025	0.4075	0.3350	0.4700
E5 zero-shot	0.1625	0.2617	0.2588	0.3200
A1 BiLSTM	0.0875	0.2025	0.1724	0.2450
A2 XLM-R probe	0.2400	0.4117	0.3591	0.4750
A3 XLM-R fine-tune	**0.3375**	**0.5142**	**0.4827**	**0.5700**
A4 PrahokBART	0.0525	0.1517	0.1035	0.1800

Recall@5: mean fraction of relevant articles recovered; **Hit@5:** any relevant article found.

MRR@10: reciprocal rank of the first relevant article, averaged over questions.

Source: *results/metrics/summary.csv. BM25 and zero-shot E5 are additional baselines.*

Interesting Discovery

The same model got 99% on titles, but 51% on real questions

TITLE-STYLE QUERIES

148 held-out titles

99.32%

NATURAL KHMER QUESTIONS

200 benchmark questions

51.42%

We trained mostly on legal titles, but people ask questions in very different ways.

Limitations

Even the best model still makes important mistakes

60.5%

Vocabulary mismatch

The question uses different wording from the article.

19.8%

Near-misses

The right article appeared at ranks 6–10.

17.4%

Multiple articles

A question may need more than one provision or articles.

USER FLOW

1. Ask question

User types a natural Khmer legal question

2. Select method

A3 semantic / BM25 keyword / Hybrid RRF

3. Retrieve

Search the same **1,976 articles**

4. Rank results

Return top 5 articles + scores

5. View evidence

Read article text and citations

OPTIONAL: GENERATE A GROUNDED ANSWER

Retrieved legal context

Top articles + original question

DeepSeek answer

Generates response using retrieved evidence

Citation check

Checks cited article IDs against context

Live Demo

Can we retrieve the right article from 1,976 candidates?

BENCHMARK QUESTION · TQ-CIV-009

តើនរណាខ្លះអាចស្នើសុំឱ្យតុលាការប្រកាសបើកអាណាព្យាបាលទូទៅ?

REFERENCE ANSWER

Civil Code · Article 24

BM25	A3 · Fine-tuned XLM-R
Keyword search	Semantic search

Live check: same query · top 5 · all statutes · compare Article 24 ranking.

Conclusion

What I learned from comparing four neural retrievers

01 **Pretraining helped** — The from-scratch BiLSTM struggled on a small dataset.

02 **Fine-tuning won** — A3 was the strongest model on natural questions.

03 **Data still mattered** — 99.32% title Recall@5 did not mean 99% on real questions.

Next: train with more natural Khmer questions and harder negative examples.

References

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